

An Internship Report Of Thiranex Web Development Virtual Internship

Submitted

To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 2410031618

Name of Student: ANIKET KUMAR

Batch: 2024-2028

2026-27

CANDIDATE'S DECLARATION

I, ANIKET KUMAR, do hereby declare that the internship report titled “Thiranex Web Development Virtual Internship” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: ANIKET KUMAR

Roll No.: 2410031618

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I would like to thank the Thiranex Education LLP team for providing me the opportunity and the structured platform to learn and gain hands-on exposure in the domain of Web Development. I am also thankful to my institute, IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement. I express my gratitude to my parents for their blessings.

Signature:-

Student Name: ANIKET KUMAR

Roll No.: 2410031618

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Certificate

O F A C H I E V E M E N T

This acknowledgment is proudly presented to

Aniket kumar

This is to certify that **Aniket kumar** has successfully completed an internship in
Web Development from **30 Jul 2026** to **29 Aug 2026**.



VERIFIED CERTIFICATE
ID: THX-JUL3126-035



Hariharan M
Founder & CEO



Date: 30 July 2026

TO,

Aniket kumar

Intern ID: THX-JUL3126-035

Subject: Offer Letter for Internship in Web Development

Dear **Aniket kumar**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

Internship Role	Intern - Web Development
Commencement Date	30 Jul 2026
Completion Date	29 Aug 2026
Work Mode	Remote / Project-Based

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.
For verification, contact Thiranex Verification Cell.
www.thiranex.in



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Virtual Internship undertaken with Thiranex Education LLP, in the domain of “Web Development”. The internship was carried out as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. The primary aim of this internship was to build a strong, practical foundation in modern front-end web development — semantic HTML5 and accessibility, responsive CSS architecture, asynchronous JavaScript with REST APIs, and full-stack project deployment — and to apply it to a real, deployed, multi-page web application.

The internship followed a structured, task-based curriculum combining conceptual learning with hands-on project work, culminating in a live, publicly deployed portfolio website that itself hosts two additional working applications built during the internship. On successful completion of all the requirements, a Virtual Internship Certificate was awarded by Thiranex Education LLP.

4.2 Organization Profile

Thiranex Education LLP is an MSME-registered virtual internship and skill development organization, officially registered under the Ministry of Micro, Small and Medium Enterprises (MSME), Government of India. Thiranex provides 100% free, verified virtual internships across 15+ domains, including Web Development, AI/ML, Data Science, Python, and Cybersecurity, with a curriculum engineered to bridge academic learning with production-level industry standards. Students complete real, project-based deliverables and receive a verified completion certificate on successful evaluation.

4.3 Problem Statement

Although engineering students commonly learn HTML, CSS, and JavaScript as part of their academic coursework, very few get structured, hands-on exposure to building a complete, production-style web application — one that is accessible, responsive across devices, consumes live data from a public API, and is actually deployed to a public URL. Concepts such as semantic markup, WCAG accessibility, mobile-first CSS architecture, and client-side routing are often understood only in isolation. The problem addressed by this internship was to practically apply all of these concepts together, by designing and shipping one real, deployed, multi-page web application rather than a set of disconnected exercises.

4.4 Project Objectives

- To restructure a portfolio website using semantic HTML5 and WCAG-aligned accessibility practices.
- To design a fully responsive, mobile-first layout using CSS Grid, Flexbox, and custom properties.
- To implement a persisted light/dark theme driven entirely by CSS custom properties.
- To fetch and render live data from a public REST API using the Fetch API and `async/await`.
- To implement comprehensive error handling for network failures and invalid user input.

- To architect a modular, multi-view front-end application with client-side routing.
- To deploy the completed application to a live, publicly accessible URL.

4.5 Scope of the Project

The scope of this internship spans the complete front-end web development workflow — starting from semantic, accessible HTML structure, through responsive CSS architecture and theming, to consuming a live REST API asynchronously, and finally to architecting and deploying a modular, multi-view e-commerce-style application with client-side routing. The internship was limited to client-side web technologies; no backend server or database was built, and API data was sourced from a free, public, keyless REST API.

4.6 Technologies and Tools Used

Category	Tools / Technologies
Languages	HTML5, CSS3, JavaScript (ES2020+)
Architecture	ES Modules, component-based CSS, hash-based client-side routing
APIs	Fetch API, async/await, Open-Meteo REST API (weather & geocoding)
Accessibility	ARIA roles/labels, semantic landmarks, keyboard navigation, WCAG contrast
Tooling	Git, GitHub, GitHub Pages / Netlify, Terser & clean-css (asset minification)

4.7 System Architecture

- **Presentation Layer:** Semantic HTML5 templates for each view, styled with a mobile-first, token-driven CSS system (custom properties for colour, type, and spacing).
- **Routing Layer:** A lightweight hash-based client-side router matches URL patterns (including dynamic parameters) to view-render functions, with no server-side rewrite rules required.
- **Data Layer:** A static product catalog module and, separately, a live REST API client (Fetch + async/await) that resolves a searched city and fetches current/forecast weather data.
- **State Layer:** A small publish/subscribe store manages cart state independently of the views that read and write it.
- **View Layer:** One module per screen (home, product detail, cart, checkout), each owning only its own DOM and event listeners.
- **Deployment Layer:** Static hosting via GitHub Pages / Netlify, with a minified production build (Terser for JavaScript, clean-css for CSS).

4.8 Methodology

The internship followed a structured, task-based methodology, with each task building directly on the deliverable before it, as summarized in the table below.

Task	Module	Description
1	HTML5 Semantic Structure & Accessibility	Rebuilt the portfolio as a multi-page site using semantic landmarks, ARIA labelling, SEO-friendly meta tags, and a fully tab-navigable, accessible contact form.
2	Advanced CSS3 & Responsive Architecture	Converted the layout to a mobile-first system using CSS Grid and Flexbox, and introduced CSS custom properties to drive a persisted light/dark theme and a two-dimensional bento-style grid layout.
3	Asynchronous JavaScript & RESTful APIs	Built a real-time Weather Dashboard using the Fetch API and async/await, with comprehensive error handling, nested-JSON rendering, and a city search with disambiguation.
4	Full-Stack Deployment & Project Architecture	Combined all prior modules into a capstone e-commerce product catalog with a modular ES-module architecture, client-side routing, a persistent cart, a minified production build, and live deployment.

Note: There was no stipend associated with this internship.

4.9 Expected Outcomes

- Ability to write accessible, semantic HTML5 that satisfies WCAG-oriented checks.
- Practical understanding of responsive, mobile-first CSS architecture, including Grid, Flexbox, and custom-property-driven theming.
- Hands-on exposure to fetching and rendering live REST API data with robust error handling.
- Capability to architect a modular front-end codebase and implement client-side routing without a framework.
- Skill to deploy a complete web application to a live, publicly accessible URL, including diagnosing and fixing real deployment issues.
- A completed, live portfolio website hosting all deliverables, submitted as the outcome of the internship.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate issued by Thiranex Education LLP, confirming successful completion of the Web Development Virtual Internship, is attached in Chapter 3 of this report.

5. BIBLIOGRAPHY/REFERENCES

- 1 Thiranex Education LLP, “Web Development Virtual Internship,” Certificate of Completion, 2026.
- 2 MDN Web Docs — HTML: A good basis for accessibility, <https://developer.mozilla.org/en-US/docs/Learn/Accessibility/HTML>
- 3 MDN Web Docs — CSS Grid Layout, https://developer.mozilla.org/en-US/docs/Web/CSS/CSS_grid_layout
- 4 MDN Web Docs — Using the Fetch API, https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API/Using_Fetch
- 5 Open-Meteo — Free Weather API, <https://open-meteo.com/>
- 6 W3C Web Accessibility Initiative (WAI), <https://www.w3.org/WAI/>
- 7 GitHub Pages Documentation, <https://docs.github.com/en/pages>
- 8 Netlify Documentation, <https://docs.netlify.com/>